

Are Commonly Practiced Trip Reduction Strategies Congruent to any Context?

A Perspective from Bangladesh

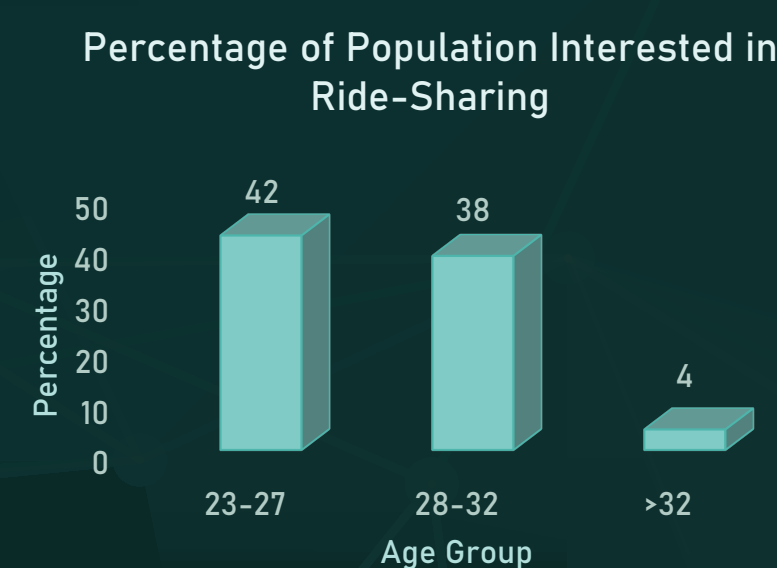
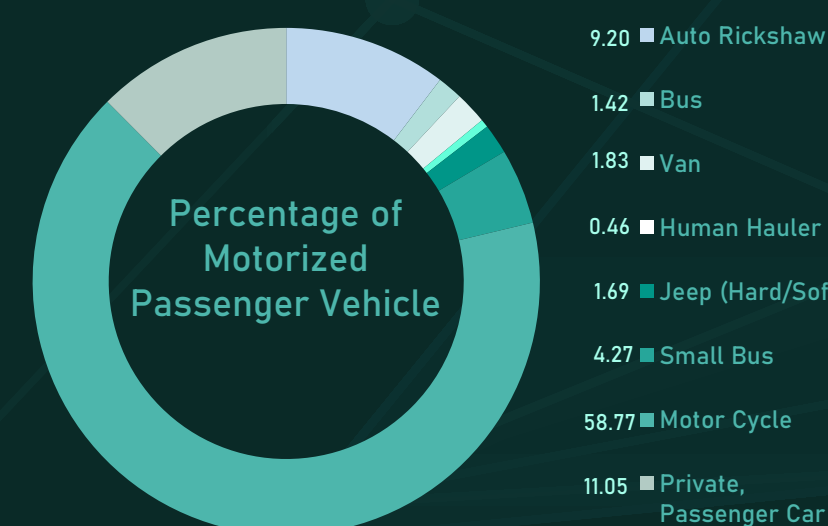
Introduction

Reducing private vehicle-based trips is supposedly one of the first strategies considered in the process of sustainable transport planning. Traditionally, most common strategies to reduce these trips are cost increment, restraint pricing, tax imposition, compact design, increasing accessibility to public transport etc. Whereas, recent studies have shown that, modal selection is rather highly dependent on the intangible variables like comfort, safety, feeling of status etc., than tangibles variables like cost, trip distance, accessibility to public transport etc. By analyzing several research on travel mode selection in Bangladesh, this study aims to find if those commonly practiced strategies are globally congruent or in different context from where most of the studies are conducted; like in Bangladesh, those strategies will become less effective.

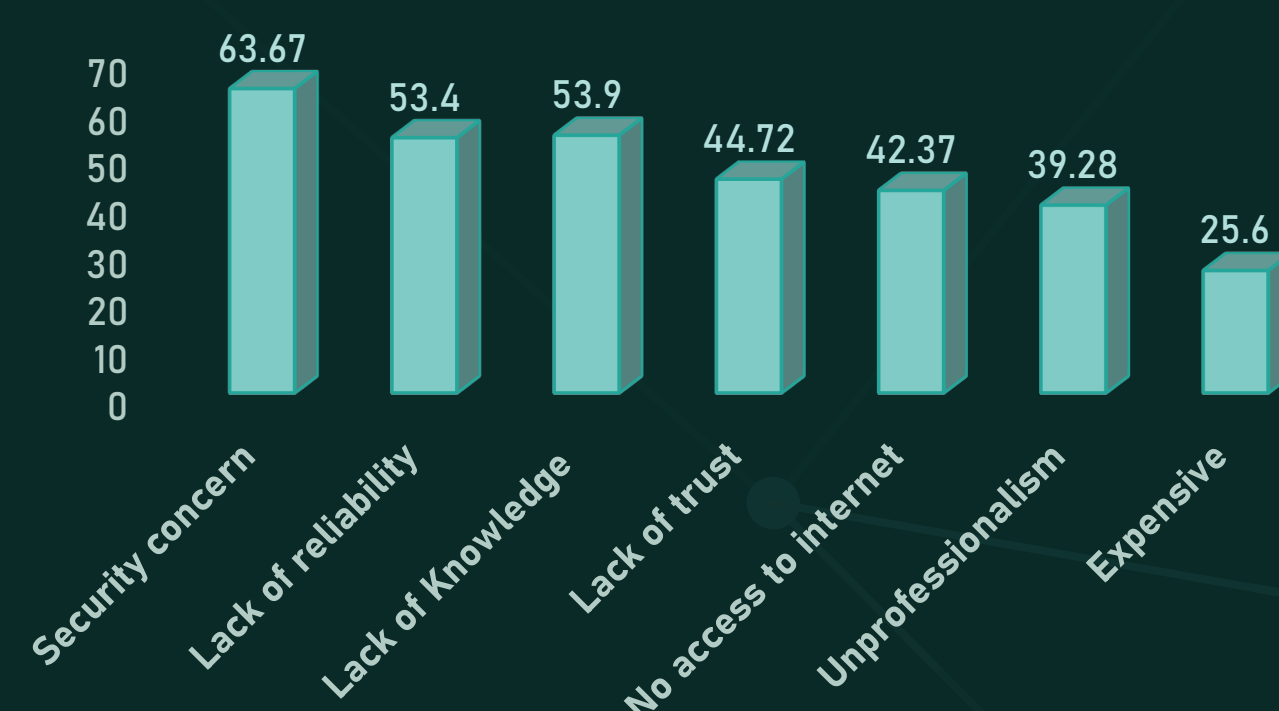
Context in Bangladesh

- › Limited number of public transport (bus)
- › Deprecation in using public transport (unskilled bus drivers, accidents, theft, robbery)
- › Excessive tax on cars failed to discourage people in buying private vehicles.
- › Lack of parking spaces
- › Absence of pricing control.
- › Evoking awareness of cycling
- › Emerging popularity of ride sharing services
- › Lack of transport infrastructure
- › Absence of a mass transit system
- › Inadequate per capita public transport share
- › Private sector can involve in public transport sector
- › Excessive usage of motorbike
- › Alarmingly low traffic speed
- › Annual new private vehicle ownerships are on decline

Adhikary (2022), Andaleeb et al. (2007), Katz & Rahman (2010)



Factors behind Avoiding Ride-sharing Services



Syed Mafiz Kamal & Noor A Ahsan (2018)

Commonly Practiced Strategies Around the World

Pricing/Taxation

- › Road Pricing
- › Fuel Pricing
- › Parking Charges
- › Information Pricing
- › Car ownership Taation
- › Pollution Pricing
- › Commuted Payments

Land Use Planning

- › Urban Concentration
- › Discouraging Peripheral Developments
- › Mix Use Development
- › Design of new developments to facilitate walking/cycling
- › Design of new developments to provide access to transport services

Physical measures

- › Public transport priority
- › Public transport roadspace
- › Cycle priority
- › Cycle roadspace
- › Pedestrian priority

Capacity Management

- › Freeze on New Road Infrastructure
- › Road Capacity Throttling
- › Integrated Planning

Policy Measures

- › Public transport deregulation
- › Company car policy
- › Lift sharing/car pooling schemes
- › Encourage opportunities for intermodality

Subsidies

- › Car pooling subsidies
- › Cycle subsidy/investment in cycle networks
- › Walk subsidy/investment in walkways
- › Public transport subsidy—operators

Communication & Technology

- › Teleworking/ telecommuting
- › Teleshopping/telebanking
- › Telematics/informatics—providing information remotely
- › Telematics—route planning and electronic guidance

Ride Sharing

- › Reducing pricing
- › Government approval
- › Making user interface more comprehensive

Marshall & Banister (1997)

Context Comparison from Bangladesh View

- › Road infrastructure is not even close to enough for capacity management.
- › Parking complexities induces more trips rather than reducing because of using the drivers
- › Ride sharing service is being the alternative of private car rather than reducing trips.
- › Teleworking forged to non-trip-based employment.
- › Safety, comfort, mentality of showing off creates congestion in compact residential area.
- › New mass rail and routes can decline pressure on roads.
- › Public transport system is not built up on monopoly.
- › Government have to subsidize fuel rather than imposing extra tax on fuel.
- › Private cars are sometimes seen as a safety shield from unwanted experiences

Appropriate Strategies for Bangladesh

- › Fostering safer condition for pedestrian and cycling users and encouraging active transport for shorter distance travel
- › Increasing railway efficiency and providing more transit alternatives
- › Improving institutional structure and reducing corruption from authorities who provide road permits and collects tax

Conclusion

After comparing Bangladesh context with the contexts of the developed countries, from where most of the strategies are induced; it is clear that, some of those commonly practiced strategies are verimuch context specific and needs to be assessed and adjusted before adopting in other contexts.